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BSCS 4A

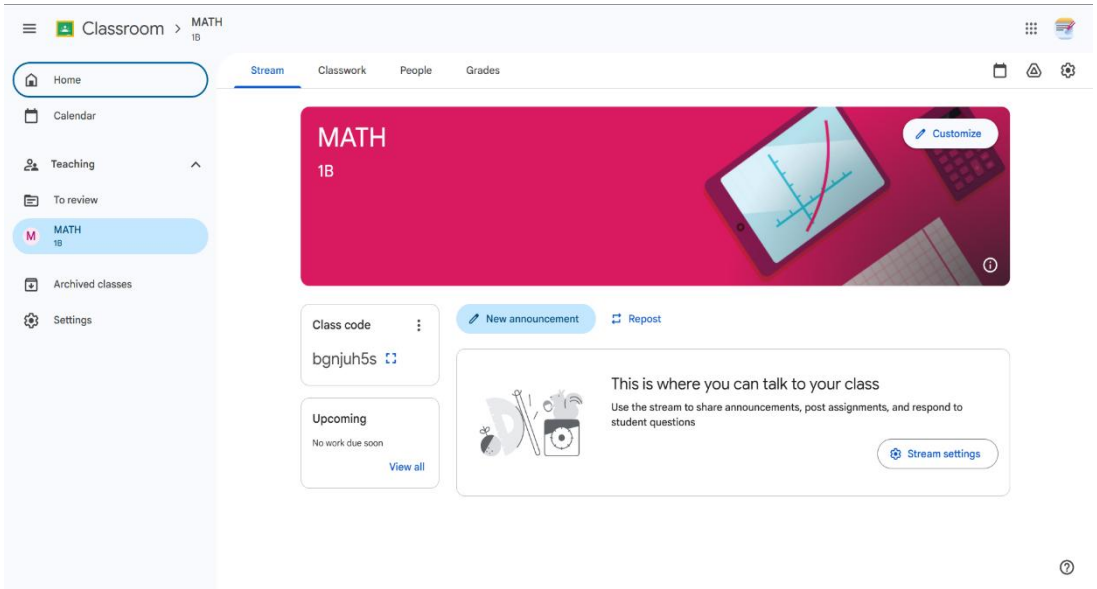
ITEC 80B - HCI
Final Project/Final Exam

Final Project: Human-Centered Interface Analysis and Redesign

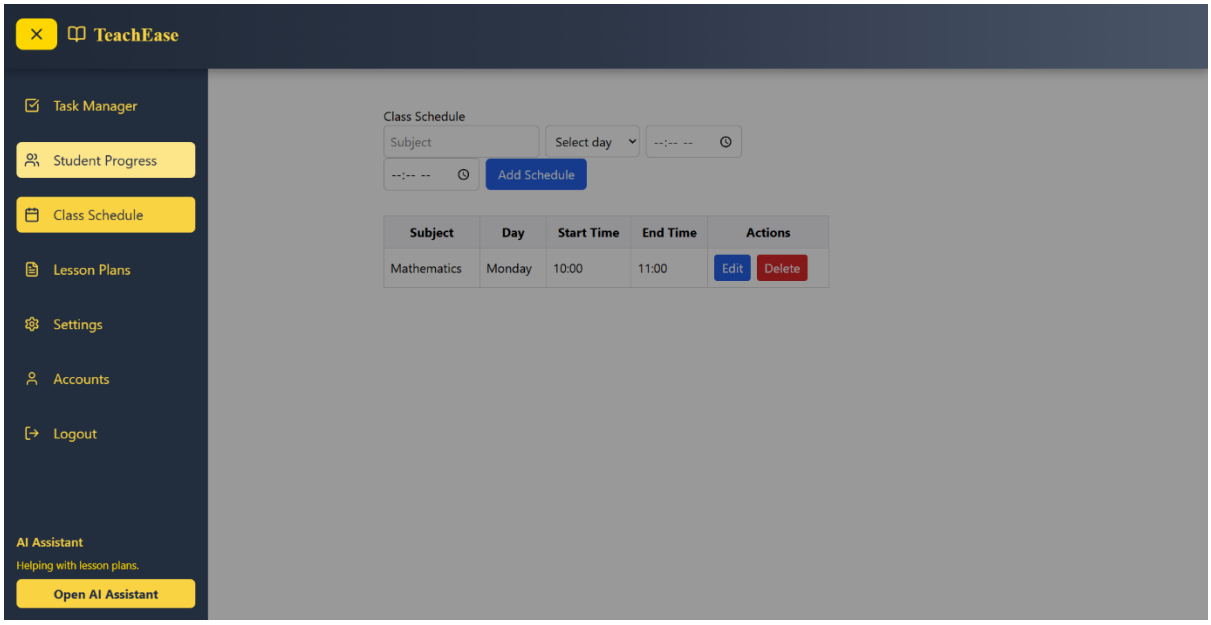
Step 1: Chosen Interface

The chosen interface for this project is the google classroom, a popular app for student and teacher in philippines

Step 2: Interface Analysis



Google Classroom (Teacher Interface)
So, Google Classroom is basically an online tool that helps teachers keep track of all their students, classwork and grades. It has a nice clean design, so you know exactly where everything is at a glance. The navigation bar is on the left side of the screen with four tabs across the top (Stream, Classwork, People and Grades) that are easy to find. Teachers use this platform to create and send assignments, view their students' submission status, add/update class materials, alert students of any changes/updates and keep track of their grades. Although this platform was built with user-friendliness in mind, it can sometimes be difficult for new users to find their way around.



TeachEase (Teacher Interface)

TeachEase is an academic management interface aimed at teachers. The main purpose of the TeachEase dashboard is to give teachers a central place to see how all their students are performing in their classes and what tasks are being completed in their classrooms. The navigation pane of the dashboard has several different categories of information including Task Manager, Student Progress, Class Schedule, Lesson Plans, etc. On the right side of the screen, teachers see the overall metrics for each student (e.g., number of activities completed, number of assignments submitted, overall average of student's quiz and exam scores). TeachEase is created for teachers that want to quickly assess how well each individual student is progressing academically and keep track of where that particular student needs additional assistance. The functions that teachers typically perform using TeachEase include searching for students, reviewing the academic progress of each student, preparing plans for each lesson, verifying that students are completing their assigned tasks, and utilizing the built-in AI assistant to assist with planning lessons and making instructional choices. TeachEase provides a user-friendly interface.

Step 3: Human Factors and Cognitive Analysis

TeachEase was designed to provide a clear, easy-to-read interface for teachers to access the cognitive aspects of their jobs. By using simple visual indicators, such as coloured icons, colour-coded progress indicators and card-style layouts for easy retrieval of all academic data about students, teachers are able to quickly assess the academic level of students and any issues they may be having without spending time reading through long, complicated reports.

TeachEase finds ways to reduce the amount of mental work you do to keep track of all the stuff related to teaching. In addition to grouping features together according to what they relate to, (e.g., Assignments, Activities, Quizzes, etc.), TeachEase has a consistent sidebar area for the features. Therefore, there is no need for teachers to look for everything repeatedly, so it's easy to get into the flow of what you're doing. Teachers can also use the built-in AI assistant to help develop lesson plans. This means that a teacher can get ideas for lesson plans without having to create them from scratch, leaving more time for refining their instructional materials.

TeachEase helps teachers learn about the system through recognization instead of working from memory, and by clearly labeling the menus, using large easy to read icons, and giving teachers guided prompts through AI while still allowing them to remember what they need to do with their own memory. Looking at it from the Mental Model perspective, the TeachEase software interface fits in with a real classroom by connecting the analysis of student performance to the AI supported lesson planing. The user interface has good ergonomic design elements like well sized readable fonts, spacing for users' eyes, and easy access to user controls for ease of usage over long peroiods of time. Overall, TeachEase combines good usability and cognitive support, which are established through it's intuitive design and AI support funcionalility.

Step 4: Analysis Table

Aspect	Observation	Issue Identified	Effect On User
Human Factors / Cognitive	Top tabs and sidebar navigation organize classes, assignments, and students.	Navigation has less persistent; top tabs require switching screens.	Users may take longer to find sections, increasing cognitive load.
Human Factors / Cognitive	Student info (grades, assignments, progress) shown separately per assignment; no single-dashboard overview.	Requires multiple clicks to see overall progress.	Teachers spend more time scanning and aggregating information.
Interaction Principles (Affordances)	Buttons in **Create Assignment**, **Grade**, **View**, **Return** are standard, with	AI-assisted features are missing.	Users may overlook some actions; less intuitive for new users.

	minimal color distinction.		
Interaction Principles (Feedback)	Assignment completion shown as percentages; no combined numeric + color-coded indicators.	Limited visual feedback.	Teachers will need to open multiple screens to view student progress.
Interaction Principles (Visibility)	Only relevant info visible per tab (students, grades, or assignments).	Cannot see all key data in one screen.	Users will navigate between tabs, slowing workflow.
Interaction Principles (Constraints)	Fields like class name and assignment details are locked; can only edit via specific buttons.	No clear visual distinction between editable vs non-editable fields.	May confuse users who try to edit directly.
Interface Guidelines (Consistency)	Layout and typography consistent across screens.	None	Predictable UI helps reduce learning curve.
Interface Guidelines (Clarity)	Assignments show titles and due dates; action buttons present.	None	Scanning many assignments can take longer.
Interface Guidelines (Error Prevention)	Deleting or returning assignments usually requires confirmation dialog.	None major; prevents accidental actions.	Reduces errors; users feel safer performing actions.
Minimalist design; modular layout.	Minimalist design; modular layout.	Requires multiple clicks to see overall class performance.	Teachers may spend extra time navigating multiple tabs.

Step 5: Identification of Usability Problems

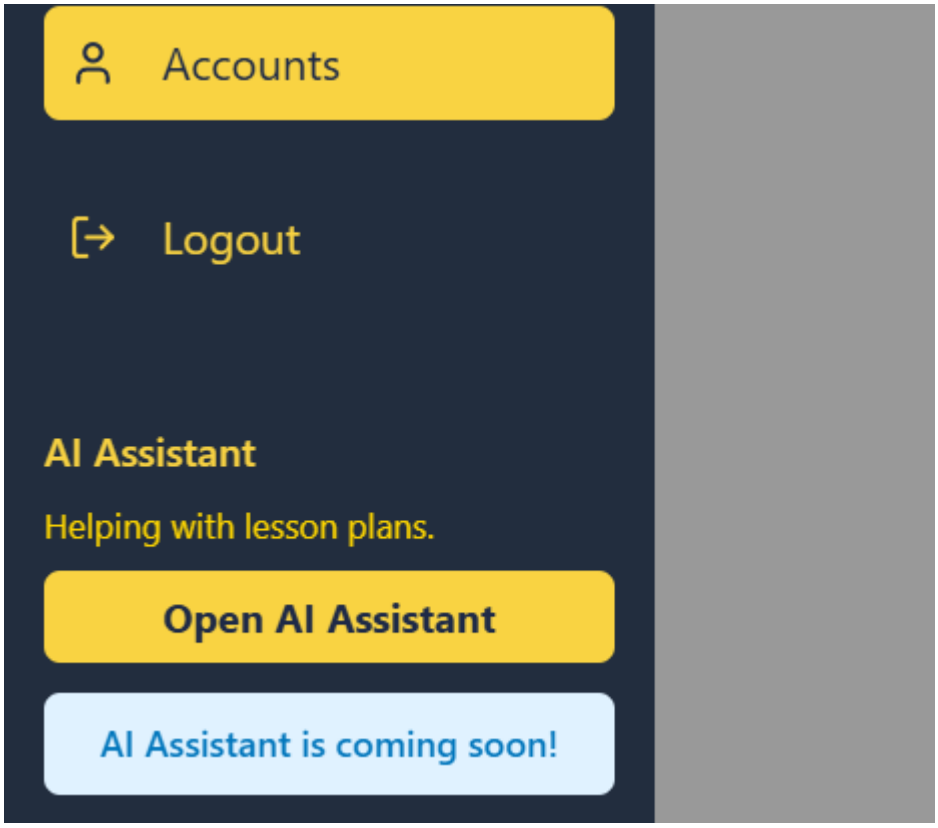
1 **High Cognitive Load in Content Organization** - Use of Virtual Classrooms Hinders Learning, Teacher Engagement, & Managing Student Engagement Teacher engagement increases when Virtual Classrooms provide the ability to see and communicate with students. Extended student engagement lowers student dropouts and enhances student

success. Teachers are the most engaged participants in educational technology and have a interest in the success and retention of their students. Virtual Classrooms offer a view of offline content and provide the opportunity for collaborative online access, and when Virtual Classrooms provide an opportunity for collaboration between teachers and students, this has a positive impact on teacher engagement and student retention. Virtual Classrooms also provide an opportunity for teachers to create supplemental course content, including video and audio lectures. Through the use of Virtual Classrooms, a teacher is able to increase their own engagement and engagement of their students because the benefits of using an educational technology (such as teacher engagement, student active, and teacher collaboration).

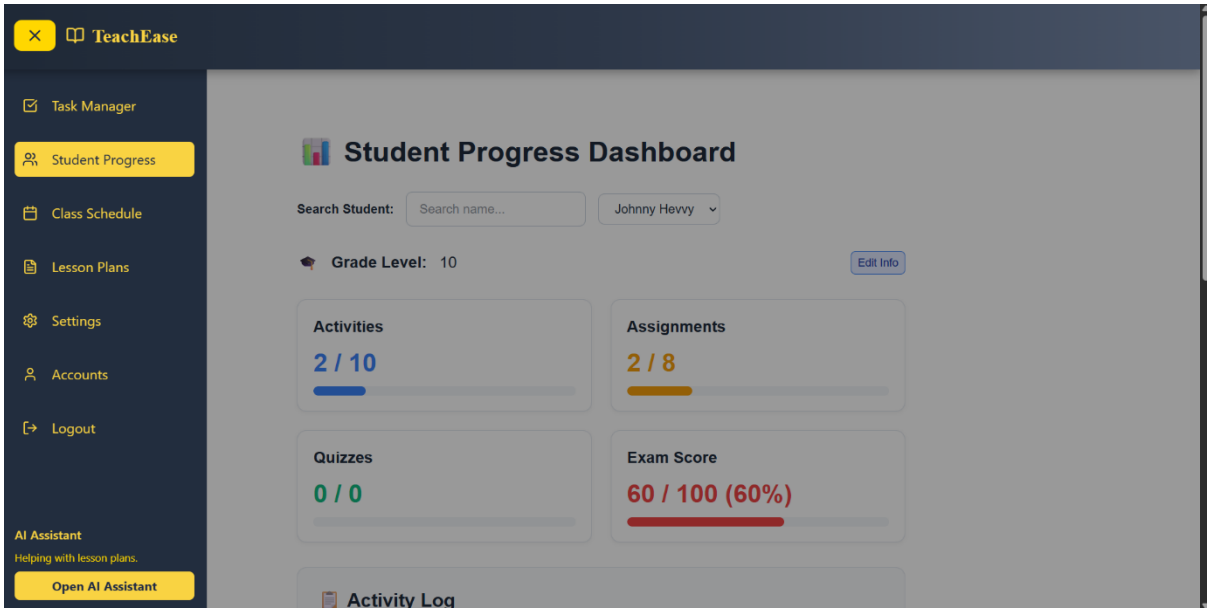
2. **Absence of Intelligent Automation for Lesson Planning -** As for lacking intelligent automations for lesson planning, teachers need integrated AI help when it comes to coming up with lesson plans or putting together files or organizing content, but since the platform is a "static container," that is the issue. In other words, teachers have to constantly hop back and forth between external AI tools, i.e., like the classroom and different apps, which causes a fragmented workflow, which is what a modern educator wants to avoid too many administrative burdens. From the HCI perspective, it violates the principle of 'minimal action', in that it requires a user to take too many steps to get to their end goal. Essentially, by not providing "accelerators" or smart automations, the system creates a high interaction cost on the end user for things that could otherwise be handled with relative ease by modern software environments.

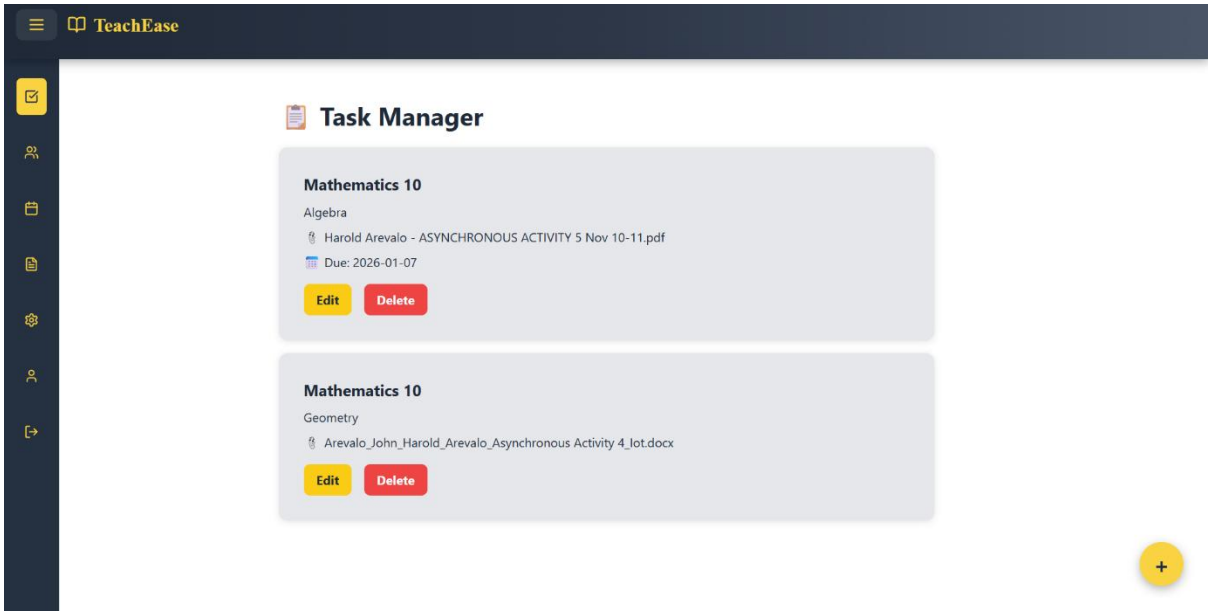
3. **Poor Aesthetic-Usability Effect** - The third major issue is that the user interface (UI) is just really boring and way too clinical. Instead of using color-coded categories, distinct icons, the design relies on this empty space with just loads of white space. That's because the platform is designed with a very strict minimalistic utility focus versus an emotional design focus. Users who use this platform every day are experiencing greater levels of screen fatigue as a result of that boring design which ultimately leads to less engagement. This violates what is known as the Aesthetic-Usability Effect in Human-Computer Interaction (HCI) where a user will perceive a more attractive interface as more intuitive and easier to use. Additionally, this violates the principle of Visual Hierarchy because it has no clear design elements that allow users to scan the screen and quickly identify the most important information.

Step 6: Redesign and Justification

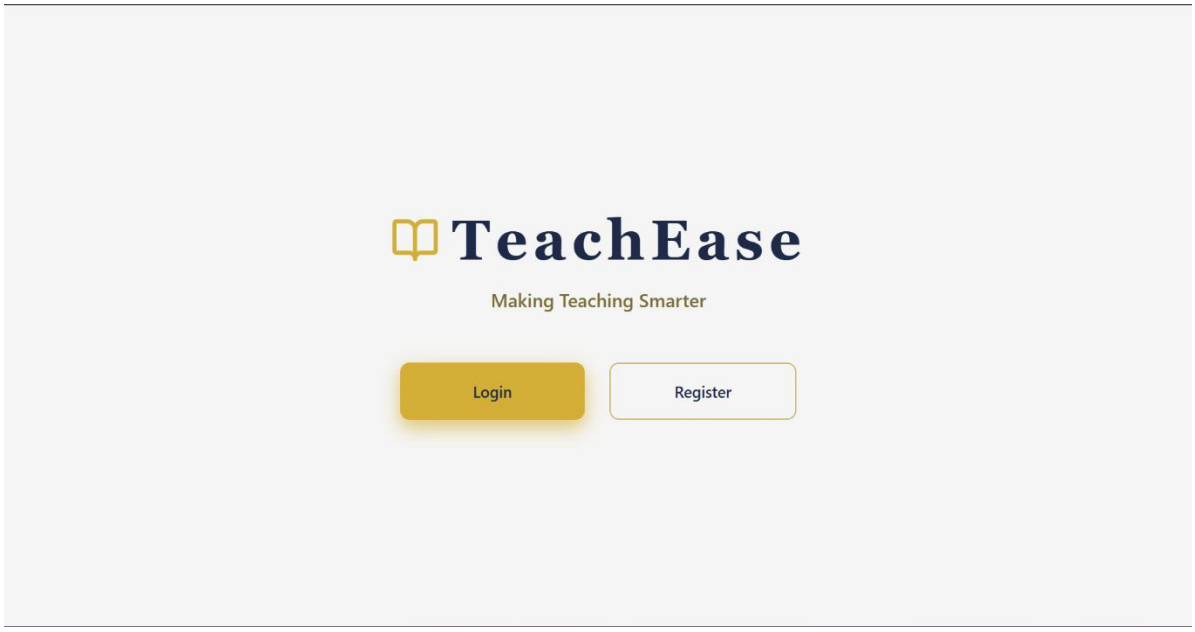


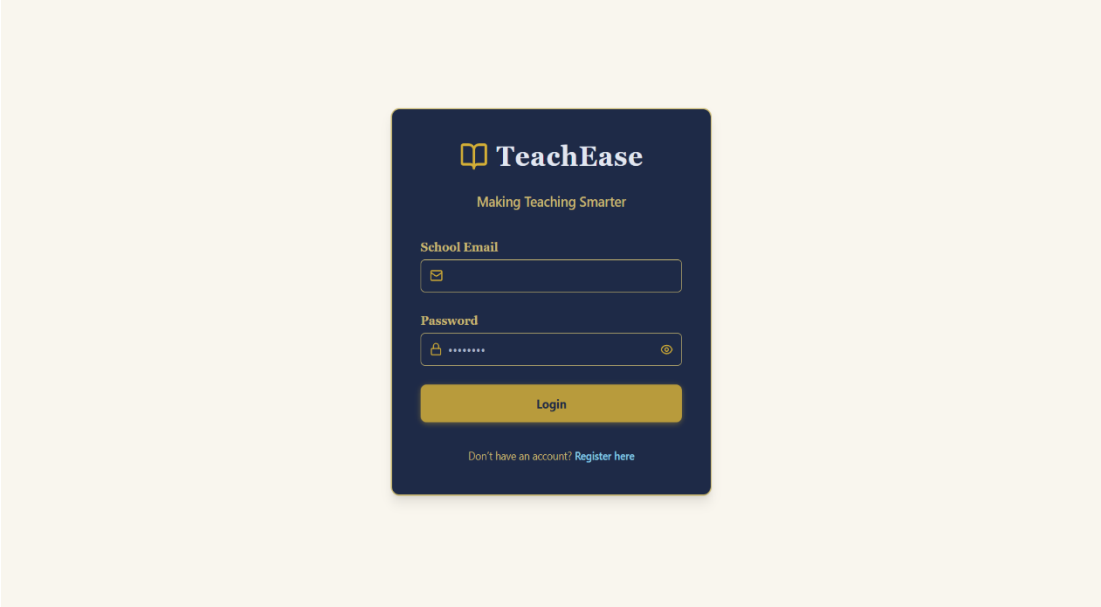
1. **AI Assistant Redesigned –** I add an open AI Assistant for better assistant when it comes in organizing teacher's lesson plans. Automate organizing their activity also help the teacher, I creating a lesson plans and activity for their students as you can see as soon as possible ai assistant will be implemented. It helps teachers to double the times of creating time of lessons.





2. **UI Simplicity Redesigned** – Make it simple than the usual google classroom unlike google classroom it will be more complex when it comes features. it also this designed help some oldies teachers or not techy teachers to used it firmly or easily no complicated buttons. It will easily to navigate what section do you want to go or what do you want to use.





3. **Redesigned Into Appealing Design and Aesthetic –** I redesigned the theme and user interface of the page to make it more appealing to users make it exactly into the users which is teachers. I will make it simple and easily to navigate from landing page, login and to main page.

Redesign Justification Table:

Usability Problem	Redesign Feature	HCI Topic Applied	Expected Improvement
Lack of Task Automation (Manual lesson planning it takes too much time)	Automated Lesson	Efficiency of Use / Productivity	Reduces teacher workload by automating lesson creation,

			enhanced productivity.
Complex Navigation (Too many buttons for non-techy teachers)	**UI Simplicity Redesign with a proper navigation bar and no complicated hidden menus**	**User Diversity / Accessibility**	**Improved Learnability for older or non-tech-savvy teachers, allowing to easily navigation.**
Unappealing/confusing Interface (Boring or complex designs leads to user a fatigue)	**Aesthetic UI & Theme Redesign with teacher-centric visual design from landing to main page.**	**Aesthetic-Usability Effect**	**Higher user engagement and satisfaction, users will mention that the system will more appealing because it is visually creative and professional.**

Step 7: Accessibility and Technology Considerations

The redesign emphasizes inclusivity through user diversity by putting the "oldies," or those who aren't very techy, at center stage and making the digital tools less complicated to use than their previous versions. We've created a more straightforward user interface with a clean and high-contrast design, so that users from different backgrounds or with visual impairments can navigate easily; we've also created easy-to-follow labels for each button. This change makes it so that people do not spend their time trying to learn how to use the software and can instead put their effort into being teachers. In addition to the digital tools themselves, we have designed an AI Assistant to provide support for users who may have difficulty with traditional manual data entry or voice commands or for those who just prefer to use voice commands, providing another means for access to the system.

The redesign of this web-based platform meets the demands of the future while honoring and retaining the traditional web management methods. The platform has enough screen space to enable the use of an AI Assistant to help educators create detailed lesson plans and can easily be accessed from any size device (mobile-first). Together, the use of AI in the redesign represents our vision for creating an Intelligent Workspace (or “Smart Workplace”), where technology not only stores files but also assists teachers in “doubling their productivity” by automating the way they structure their day-to-day lesson activities. By being very simple, yet very sophisticated, the new user interface (UI) is designed to support the ever-evolving tech

landscape (AI and other) and the practical needs that all educators and learners have every day.

Step 8: Reflection

When I working this, I made sure one of the main Human-Computer Interaction (HCI) topics was Efficiency of Use by adding an AI Assistant where it could perform Lesson Plans for you without any effort on the part of a teacher. Once I started focusing on Teacher Workflow, I found that the only way a system is considered usable is if it decreases the teacher's cognitive load and saves them time. As I stated earlier, every decision I made in this project was influenced by User-Centered Design (UCD). Instead of simply adding features to my design, I focused on providing a solution to the frustrations of "non-tech-savvy" teachers. This approach allowed me to create a Simple UI Redesign so that all teachers, regardless of age, have access to a well-organized and easy-to-use navigation system. I have learned that designing a product for real users requires a substantial amount of understanding of the Aesthetic-Usability Effect, which refers to the phenomenon in which users perceive a product as being more intuitive and approachable when the overall design is appealing to them.

During creation of my design and the focus on it was to give clear feedback to the user with the use of AI through having it categorizing and organizing the user/tasks in a simple and efficient way making it easy for the user. One of the main aspects of my app was to keep it simple, and I did this by simplifying the navigation through the interface, removing unnecessary buttons that can confuse or frustrate the user, and making it much easier for users to navigate. I learned from the start of this project that the old adage of less is more applies to interface design. A clean, simple interface looks good to the user and allows them to concentrate on their work. Also, I learned that, when designing something new, you must find a happy medium between using the new technology (AI) and what the consumer wants (simple and clear). My intention was to design an app for real-life teachers, to assist them with their classroom teaching rather than adding to their workload.